

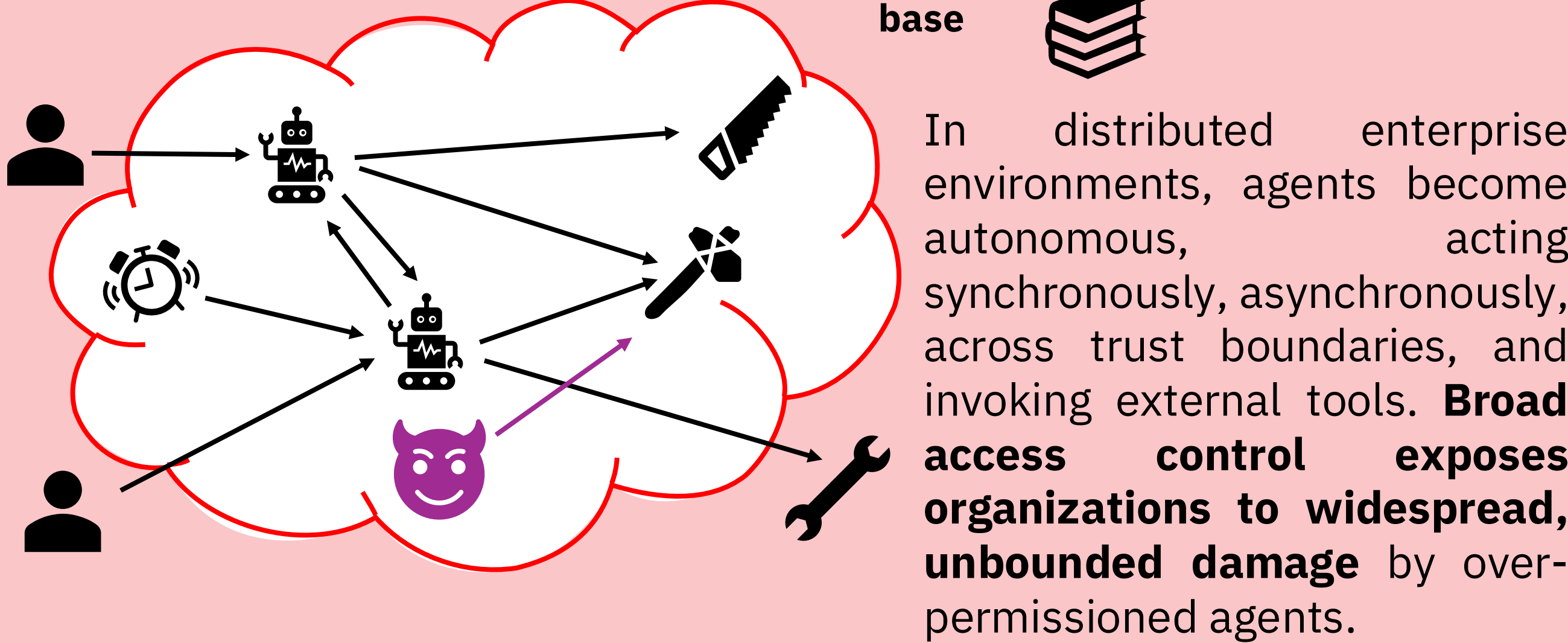
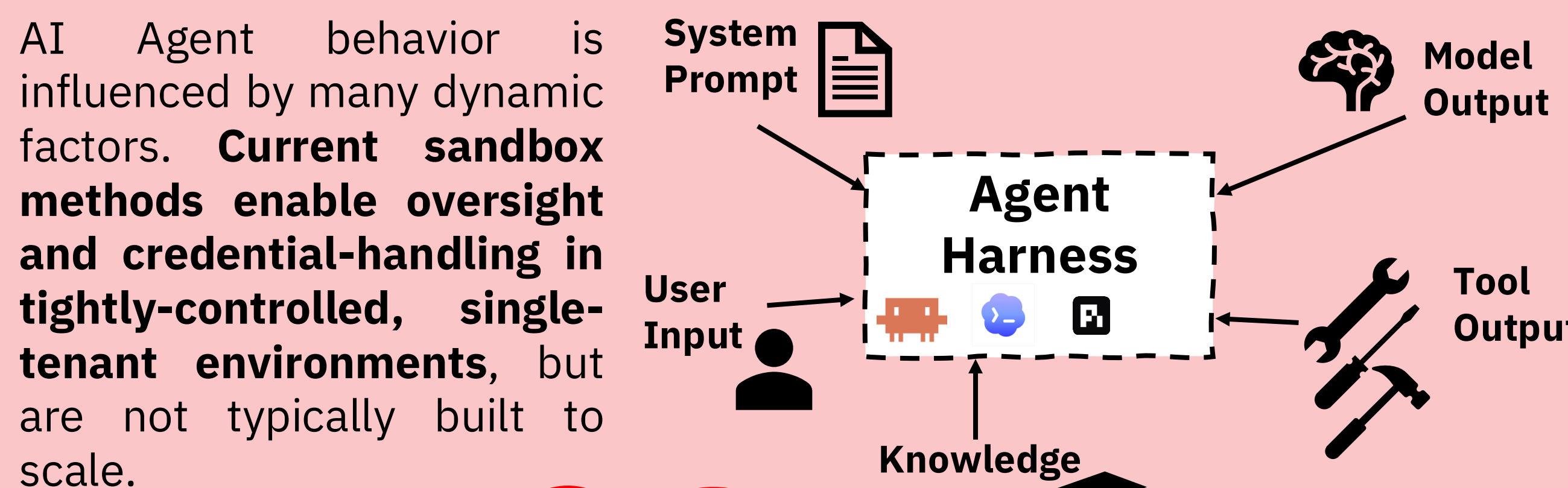
From Sandbox to Swarm: The Missing Platform Blueprint for Production-Ready Agents

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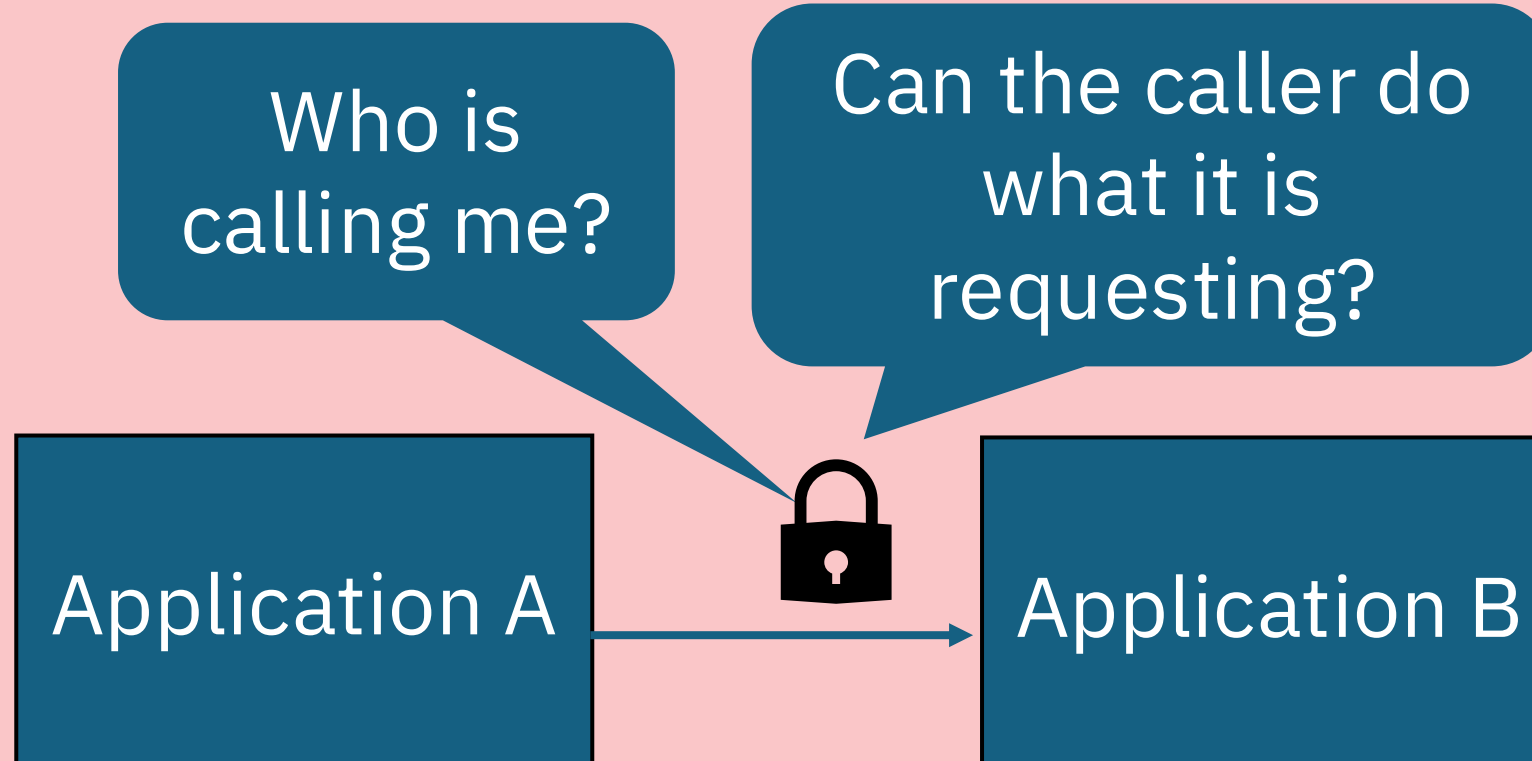
Problem

How can we establish **cryptographic chains of trust of agentic actions** to enable production deployment of highly-dynamic, non-deterministic AI agents?
Broader Implications: What **platform patterns** enable the management of AI-enabled workloads in **scaled, distributed, potentially-fragmented enterprise environments**?

Background



The Cloud Native ecosystem has been tackling the problem of distributed trust for decades. Typically, this means implementing an **authentication and authorization check at every component** upon every request – this is the principle of **Zero Trust**.

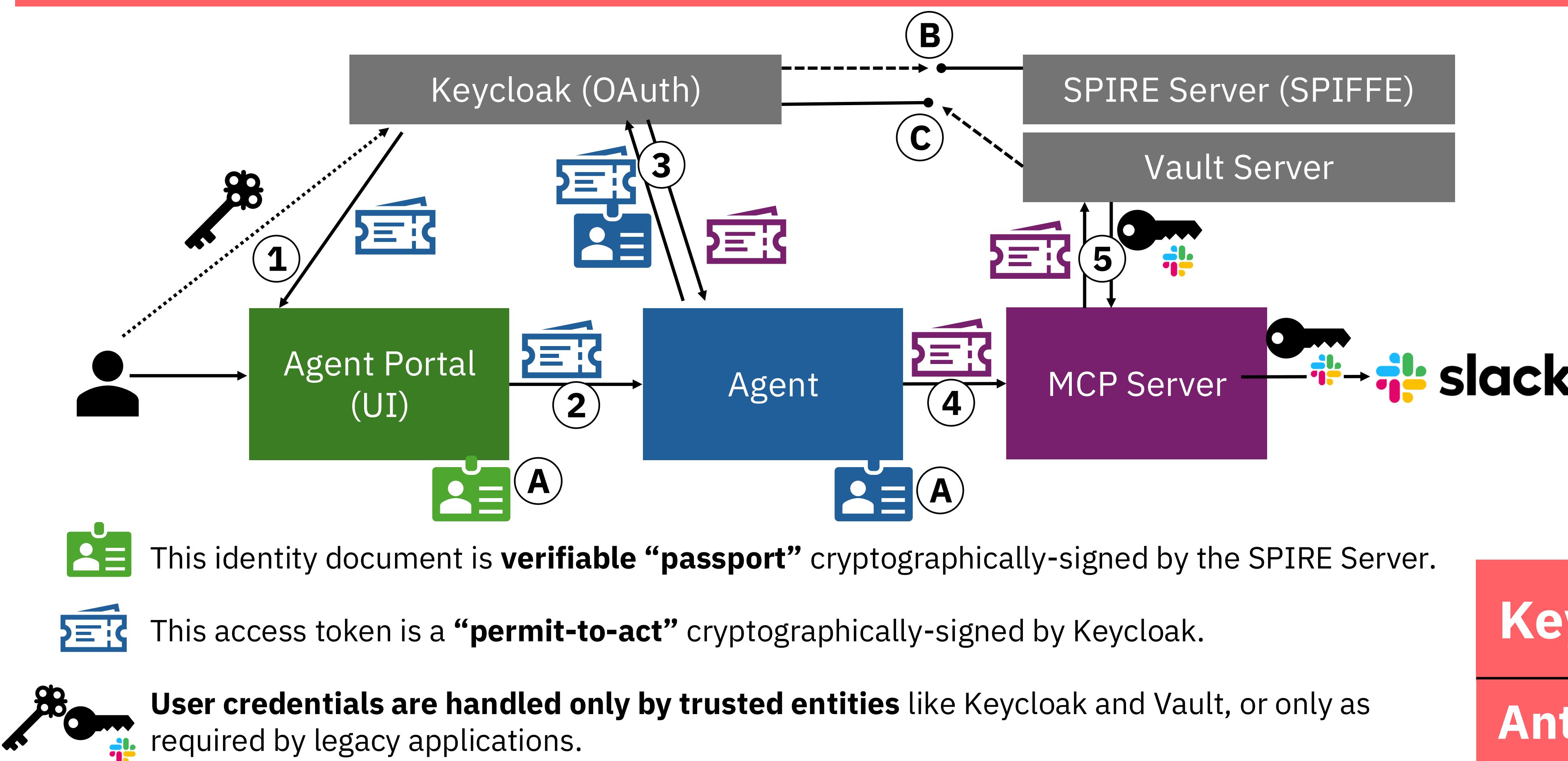


See more!

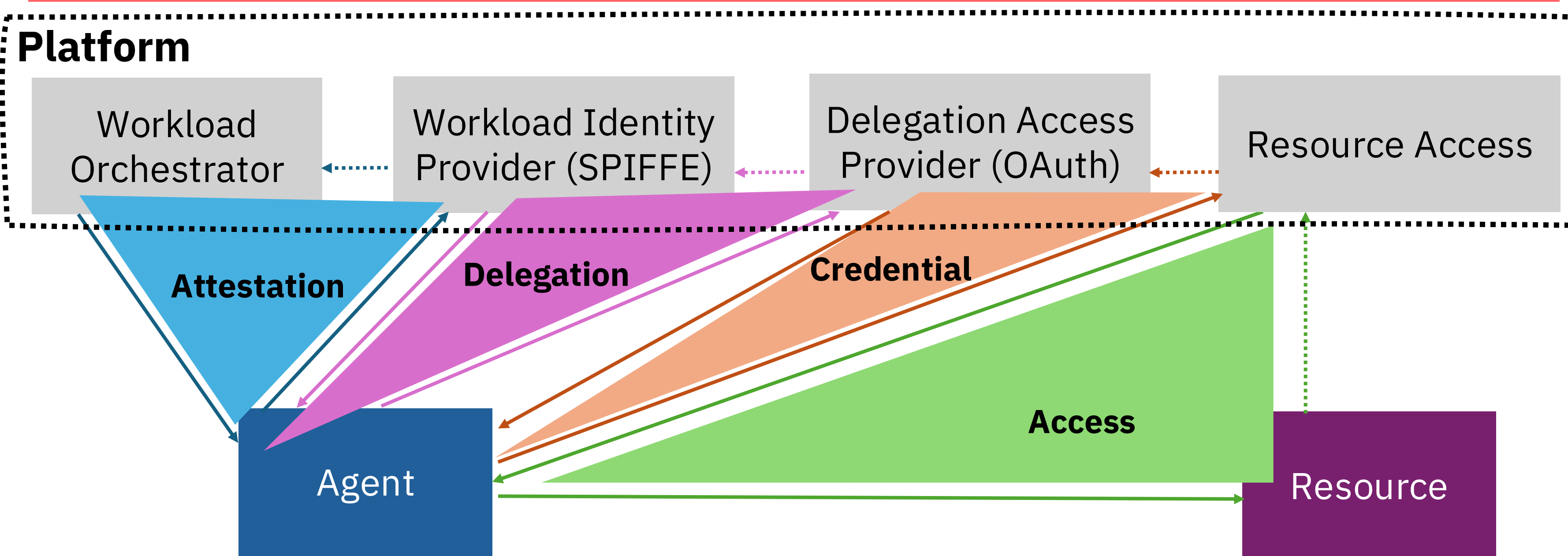


Check out our most recent work in the open-source: **rossoctl** – **dedicated to abstracting, developing, and testing platform primitives** to best enable agent security, reliability, efficiency, observability, and governance.

Demonstration of Zero Trust with a Linear Agentic Flow

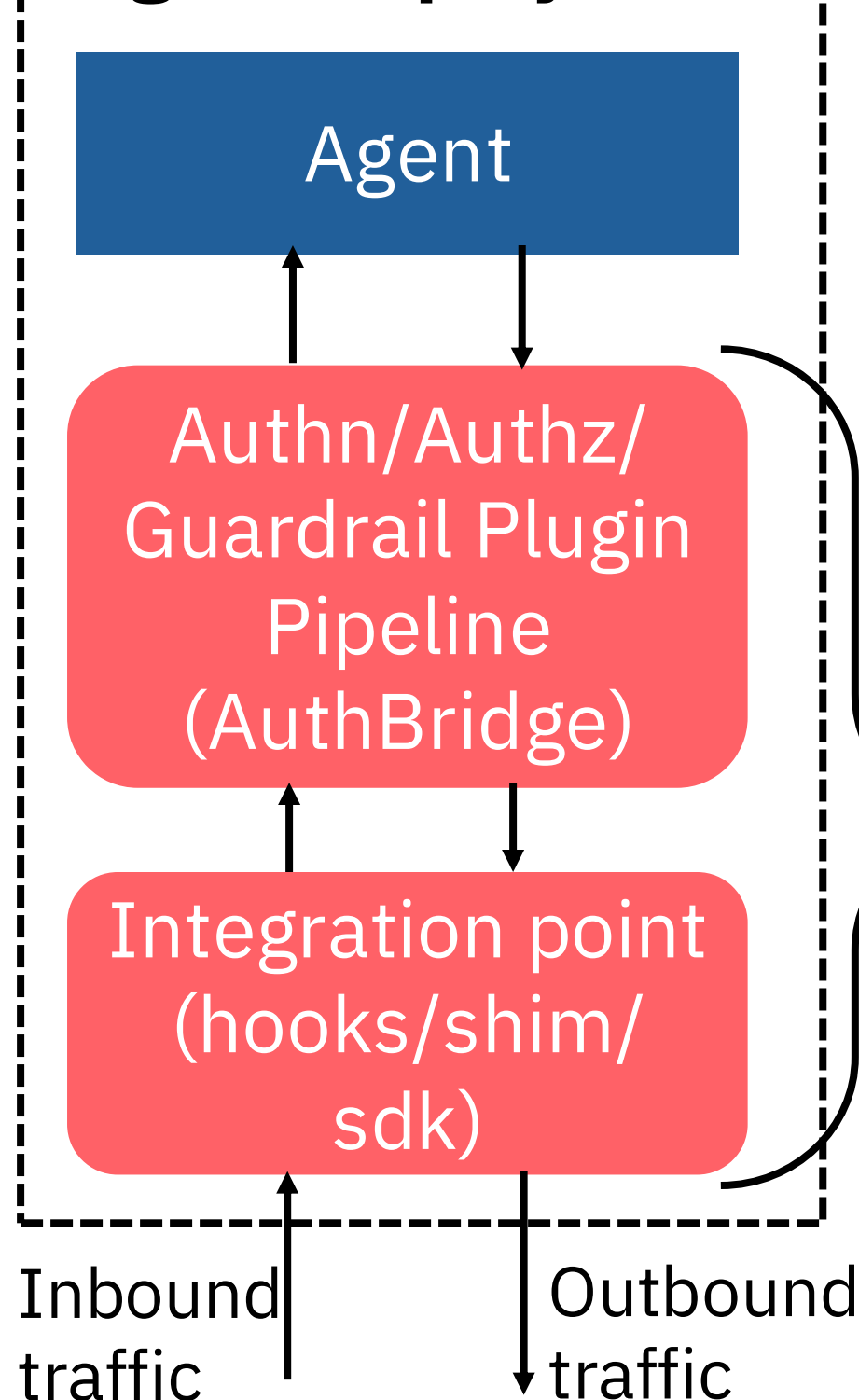


A Platform Blueprint for a Chain of Trust



From the demo, we can **abstract platform components and identify capabilities agents need** to integrate with the platform to enable strong authorization based on granular identities and cryptographic chains of trust, as indicated by the **standards-based abstractions enable interoperability** with any technologies that are standards-compliant.

Agent Deployment



Impact

We are actively **open-sourcing** implementations of the patterns developed to validate these methods.

- RossoCortex** is the dataplane that connects agents with security, context management, or observability infrastructure and tooling. This is available under **rossoctl**.
- AuthBridge** is the logical component of **RossoCortex** that handles authentication and authorization.

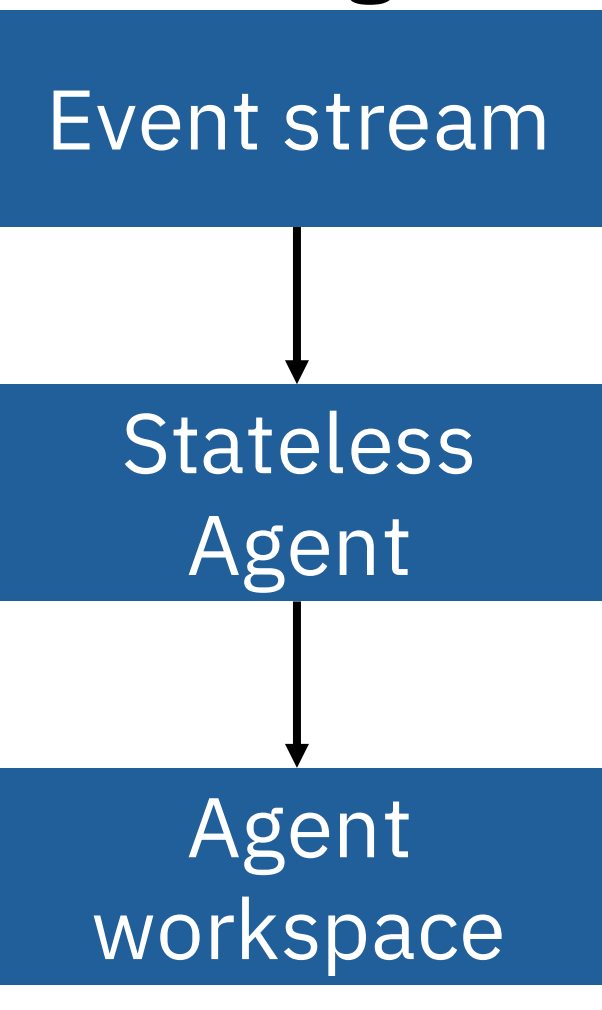
Parts of this work have been upstreamed as features in **kuadrant/mcp-gateway**, and are in-progress in **Nvidia OpenShell**.

Key Security Vulnerabilities Mitigated

Anti-Pattern	Vulnerability	Pattern
Implicit trust based on network address or static API key	Hijacked agents can impersonate other more privileged workloads	Cryptographic attestation and identity providing (SPIFFE)
Sudo-Mode service accounts for agents	Privilege-escalation of users via over-permissioned agents	Dynamic context construction via OAuth
Credential propagation	Impersonation of upstream entities by downstream entities	Credential Exchanges via OAuth Token Exchange or Vault
Static credentials stored in environment variables	Credential leakage to model services or to other entities via prompt-injection	Just-In-Time Credentialing with Vault
Framework-specific security implementations	Bypass of security or observability infrastructure	Platform-level primitives to implement security in the infrastructure

Future Work

Emerging Stateless Agent Paradigms



- Generalizing to Complex Swarms**
- Accountability tracing for multi-agent flows
 - Identity delegation for spawned sub-agents
 - End-to-end accountability tracing across dynamic
- Advanced, Dynamic Authorization**
- Intent-based access control
 - Granular data access and sharing across agents
- New deployment patterns**
- Event-driven agents
 - Primitives to improve failure recovery and resiliency
 - Token cost management and attribution